

More on the articulation of devoiced [u] in Tokyo Japanese: effects of surrounding consonants

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Abstract

Devoicing of high vowels in Tokyo Japanese is extensively studied in the literature, but one aspect of the phenomenon that still remains heavily understudied to date is the question of whether such devoiced vowels retain their lingual gesture. Since devoiced vowels are themselves inaudible, this issue should ideally be examined using a method that directly measures articulatory movements. To this end, Shaw & Kawahara (2018b) addressed this question using EMA (ElectroMagnetic Articulography), the results of which found optional but categorical deletion patterns: some vowels retain a full lingual target, just like their voiced counterparts, whereas other vowels show trajectories that are best modelled as targetless, i.e. linear interpolation between the surrounding vowels. Extending on this finding, as well as being inspired by various phonetic and phonological considerations, the current study explores the hypothesis that this probabilistic deletion of devoiced high vowels may be modulated by the manner of surrounding consonants. A new follow-up EMA-based experiment with an extended stimulus set replicates the core finding of Shaw & Kawahara (2018b) that Japanese devoiced [u] sometimes lack tongue dorsum raising gesture. The current results moreover show that surrounding consonants do indeed affect the probability of such targetless vowels, at least for some speakers. The experiment also reveals that deletion of devoiced vowels can manifest itself in different gestural coordination patterns between CC sequences and CVC sequences, although we also found an intriguing complication which goes beyond the original finding by Shaw & Kawahara (2018b).

23 1 Introduction

24 1.1 General background

25 Vowels that are adjacent to voiceless obstruents are sometimes produced with glottal
26 abduction—i.e. as voiceless—the phenomenon generally referred to as “vowel devoicing.”
27 This pattern is observed systematically across many genetically-unrelated languages, in-
28 cluding but not limited to Cheyenne (Rhodes, 1972), French (Smith, 2003), Greek (Dauer,
29 1980), Korean (Jun et al., 1998), (Andean) Spanish (Delforge, 2008), Uspanteko (Bennett,
30 2020), Uzbek (Sjoberg, 1963), and Turkish (Jannedy, 1995). Tokyo Japanese also exhibits
31 such devoicing of high vowels, and Japanese is arguably the best studied language in this
32 respect (Fujimoto 2015 for a recent review of the relevant research).

33 A general characterization of the high vowel devoicing pattern in Tokyo Japanese is
34 that high vowels are devoiced between two voiceless obstruents, as well as after a voiceless
35 obstruent and before a pause, although as we will review below, the likelihood of devoic-
36 ing is affected by various other factors, both linguistic and social. Previous studies have
37 explored this devoicing process from various perspectives, each bearing upon some im-
38 portant issues in the phonetic and phonological theory, including how devoicing is imple-
39 mented in terms of the laryngeal gesture (Fujimoto et al., 2002; Sawashima, 1971; Yosh-
40 ioka, 1981), how the precise environment affects the likelihood of devoicing (Maekawa &
41 Kikuchi, 2005; Tsuchida, 1997), its categorical or gradient nature (Nielsen, 2015; Tanner
42 et al., 2019), its interaction with lexical pitch accent (Kuriyagawa & Sawashima, 1989;
43 Maekawa, 1990; Vance, 1987) and other prosodic properties (Kilbourn-Ceron & Son-
44 deregger, 2018), its consequences (or lack thereof) for prosodic reorganization (Kondo,
45 1997, 2001; Kawahara & Shaw, 2018), its perceptual consequences (Cutler et al., 2009;
46 Ogasawara, 2013; Sugito & Hirose, 1988; Whang, 2019), its role in child-directed speech
47 (Fais et al., 2010; Martin et al., 2014), its acquisition patterns (Imaizumi & Hayashi, 1995;
48 Imaizumi et al., 1995) as well as the influence of social factors on this pattern (Imai, 2004;
49 Imaizumi et al., 1995). There is no doubt that these studies have revealed important as-
50 pects of this devoicing phenomenon in Japanese, and we understand its nature much better
51 than 50 years ago.

52 However, despite the accumulation of studies on vowel devoicing, one aspect that is
53 heavily under-addressed regarding the pattern of high vowel devoicing in Japanese—and
54 any other languages that exhibit vowel devoicing, for that matter—is the question of how
55 the lingual gesture is implemented for the devoiced vowels. This issue is directly related
56 to the question of whether these devoiced vowels are phonologically deleted or not; if the
57 high vowels are phonologically deleted, then we would expect them to lack any lingual
58 gesture. If the process at issue is phonologically devoicing rather than deletion, on the
59 other hand, we may expect that their lingual gestures are retained. Vance (2008), which
60 is the most recent and updated textbook on Japanese phonetics and phonology, indicates
61 that this issue is not settled. The current paper reports an experiment that addresses this
62 issue, observing articulatory kinematics via Electromagnetic Articulography. This paper
63 moreover tests a new, specific hypothesis that deletion probability may be modulated by
64 the surrounding consonantal environment. We will start with the overview of the relevant
65 literature on this topic, which leads us to examine this specific hypothesis.

66 **1.2 Are devoiced vowels in Japanese deleted?**

67 Since devoiced vowels lack a periodic energy source, it is difficult, if not entirely impossi-
68 ble, to infer from their acoustic profiles alone whether devoiced vowels retain their lingual
69 gestures or not. There are some studies which addressed this question via impressionistic
70 observations. Kawakami (1977) argues that vowels delete completely in some environ-
71 ments but not others, but he offers no experimental evidence for this claim. Vance (1987)
72 examines the hypothesis that devoiced high vowels in Japanese are entirely deleted but
73 ultimately rejects this hypothesis. Kondo (1997, 2001) argues that devoiced high vowels
74 are deleted based on a phonological consideration: vowel devoicing in consecutive sylla-
75 bles is often inhibited (though see Nielsen 2015), and Kondo (1997, 2001) attributes this
76 observation to a constraint against triconsonantal clusters. The underlying assumption of
77 this analysis is that devoiced vowels are deleted, resulting in consonant clusters.

78 On the other hand, Tsuchida (1997) and Kawahara (2015) point out that devoiced
79 vowels count toward a bimoraic requirement in foot-based morphophonological truncation
80 patterns (Poser, 1990), arguing that these vowels do not delete phonologically. Like these

81 two authors, Hirayama (2009) demonstrates that devoiced vowels behave just like voiced
82 vowels in the Japanese *haiku* poetry pattern, which is mora-based (Vance, 1987).

83 In line with this view, Jun and her colleagues advanced an explanation of high vowel
84 devoicing (in Korean) in terms of “gestural overlap” (Browman & Goldstein, 1992a), ac-
85 cording to which the articulatory gesture of high vowels is overlapped in time by the
86 laryngeal glottal abduction gesture of surrounding consonants (Jun & Beckman, 1993; Jun
87 et al., 1998). In this gestural overlap view, Japanese phonology does not delete the de-
88 voiced high vowels; the high vowels are merely rendered inaudible because of the glottal
89 abduction gesture that coincides in time with the vocalic gesture. This is analogous to the
90 famous case of English *perfect memory*, in which the word-final [t] in *perfect* can be made
91 inaudible due to gestural overlap with the following [m], even when the [t]’s tongue tip
92 gesture remains intact (Browman & Goldstein, 1992a). Similar to this case in English, it
93 is conceivable that lingual gestures of devoiced high vowels are present, but are merely
94 rendered inaudible because of the overlapping glottal abduction gesture. In this gestural-
95 overlap scenario, it is also possible that lingual gestures are reduced, rather than remaining
96 completely intact, assuming speakers invest less articulatory energy into sounds that are
97 difficult to perceive and hence may not contribute much to lexical retrieval (e.g. Hall et al.
98 2018; Jaeger & Buz 2018).

99 Recently we have witnessed a rise of studies addressing this question—whether de-
100 voiced high vowels are deleted or not—using instrumental techniques. Beckman and her
101 colleagues, based on the inspection of spectrograms, argue that devoiced vowels are phys-
102 ically not present (Beckman, 1982; Beckman & Shoji, 1984), suggesting that the pattern
103 should be characterized as deletion, although they also suggest that it may make sense to
104 characterize the pattern as devoicing, not deletion, from the psycholinguistic perspective;
105 i.e. Japanese speakers feel that “vowels are there” even when they are actually deleted
106 (cf. Dupoux et al. 1999, 2011; Whang 2019). Faber & Vance (2010) offer some acoustic
107 evidence for the hypothesis that vowel devoicing is best characterized as gestural overlap
108 of laryngeal gestures in Japanese (Jun & Beckman, 1993; Jun et al., 1998). Jannedy (1995)
109 and Bennett (2020) entertain a similar hypothesis for devoiced vowels in Turkish and Us-
110 panteko, respectively. Whang (2018) measured COG during devoiced vowels in Japanese
111 and argues that some devoiced vowels in Japanese are in fact deleted, while others are not.

112 More specifically, Whang (2018) argues that deletion is more likely in the environment
113 where the quality of those vowels can be recovered from surrounding consonants; e.g. af-
114 ter [ɸ], only [u]¹ is possible, while after [ʃ] both [u] and [i] are possible (see also Whang
115 2019).

116 However, generally speaking, there are limits on how much we can conclude about the
117 articulatory gestures from their resulting acoustic signals (see e.g. Browman & Goldstein
118 1989; Guenther et al. 1999; Munson et al. 2010; Perkell et al. 1993). It is thus impor-
119 tant that we address the nature of the lingual gesture of devoiced high vowels through
120 observation of articulatory movement. To that end, Matsui (2017) used EPG (Electro
121 PalatoGraphy) to examine the linguo-palatal contact pattern of devoiced syllable [s̥u], and
122 showed that the pattern remains very constant across the syllable; i.e. there does not seem
123 to be a clear change in the linguo-palatal contact pattern from [s] to [s̥u], implying that
124 the lingual gesture of the devoiced [s̥u] is absent. Nakamura (2003) on the other hand re-
125 ports that vestiges of lingual gestures of devoiced vowels can be found in his EPG data.
126 Although these two results, which seem to conflict with each other, are telling, there are
127 limits on how much we can conclude about tongue body movement—primary correlates
128 of vowel gestures (Browman & Goldstein, 1992b; Johnson et al., 1993)—from EPG data
129 in general, since EPG only registers contact with the palate. Funatsu & Fujimoto (2011)
130 used ElectroMagnetic Midsagittal Articulography (EMMA) to study articulatory gestures
131 of devoiced [i], showing that the articulatory gesture of [i] is comparable between voiced
132 [i] and devoiced [i̥]. This study however used one speaker and one pair of items (/kide/
133 vs. /kite/) with four repetitions, and offers no quantitative comparisons between the two
134 voicing conditions.

135 The most extensive study on this topic—the presence/absence of lingual gestures of
136 devoiced vowels in Japanese—to date is that of Shaw & Kawahara (2018b), who used
137 EMA (ElectroMagnetic Articulography) to study the articulatory nature of devoiced [u̥]s
138 of six naive speakers of Tokyo Japanese, and the current paper can be considered as a
139 direct follow-up of Shaw & Kawahara (2018b).

¹Here and throughout the paper, we use the IPA symbol [u] to denote the high non-front vowel in Japanese. The exact phonetic nature of this vowel, as well as how to transcribe it, is a contentious issue even in the contemporary literature (Vance, 2008). We will return to this issue in the method section (§3.2), where we justify our choice of phonetic parameters used to assess the deletion of this vowel.

Shaw & Kawahara (2018b) analyzed four dyads to compare the articulatory trajectories of CVC sequences, in which one member of each dyad contains a voiced vowel and the other a devoiced vowel. The four dyads were: (1) [ɸʊsoku] vs. [ɸuzoku], (2) [ɸʊtaisee] vs. [ɸudaika], (3) [katsutoki] vs. [katsudoo] and (4) [masʊtaa] vs. [masuda].² Their strategy, reviewed in further detail below in §3.2, is to compare the articulatory trajectory of [CuC] with respect to that of [CuC] and [CɸC], the latter of which is characterized by linear interpolation between the surrounding vowels (Choi, 1995; Cohn, 1993; Keating, 1988; Pierrehumbert & Beckman, 1988). Their conclusion in a nutshell is that some productions contain no articulatory target, while others show lingual targets that are no different from voiced vowels; i.e. they found a pattern of optional but categorical deletion. Moreover, they found some variation with respect to how often each item showed devoiced vowels without lingual targets: devoiced vowels were more likely to be targetless between [ɸ] and [t] ([ɸʊtaisee]) than between [ɸ] and [s] ([ɸʊsoku]). This asymmetry consistent across the speakers (see also discussion in 3.2).

An intriguing hypothesis that emerges from this result is that vowel deletion probability may be systematically modulated via surrounding consonant environments—Japanese [u]s may be more likely to lack lingual gesture between a fricative and a stop than between two fricatives. We expand in the next subsection why this is an interesting and plausible hypothesis to entertain, although we also note at this point that the results by Shaw & Kawahara (2018b) are based on just one dyad per each phonological condition.

1.3 The current hypothesis

The general hypothesis pursued in this study is that the probability of [u] lacking its lingual gesture—which we equate with the probability of phonological deletion for the sake of exposition here³—is modulated by surrounding consonantal environment. A more specific hypothesis is that [u]s are more likely to be phonologically deleted when surrounded by a fricative and a stop than when surrounded by two fricatives. As mentioned above, one reason to entertain this hypothesis is the results reported by Shaw & Kawahara (2018b),

²Glosses: (1) shortage vs. attachment, (2) subjectivity vs. theme song, (3) when winning vs. activity and (4) master vs. PERSONAL NAME.

³See section 6 for further discussion on this point

167 who show that targetless [u]s were more likely in [ʃutaisee] than in [ϕʊsoku]. However, it
168 is hard to know whether or not their findings are generalizable to other items with similar
169 phonological properties, because their results are based on one dyad per each phonological
170 condition.

171 Nevertheless, this hypothesis dovetails with an observation by Starr & Shih (2017),
172 who found that devoiced vowels are often skipped in the text-setting of Japanese songs,
173 and this is especially so when they are surrounded by a fricative and a stop. Their observa-
174 tion may suggest that Japanese composers are sensitive to the higher likelihood of vowel
175 deletion in this environment. The higher likelihood of deletion after a fricative and be-
176 fore a stop is also compatible with the general cross-linguistic observation about prosodic
177 wellformedness, namely, syllable contact laws (Murray & Vennemann, 1983; Vennemann,
178 1988)—languages generally prefer sonority fall to sonority plateau/rise across a syllable
179 boundary. To the extent that Japanese is also sensitive to such prosodic wellformed-
180 ness conditions (cf. Berent et al. 2007, 2008), we may expect Japanese high vowels to
181 delete more often in the environment which would result in a fricative-stop cluster than
182 a fricative-fricative cluster. To view it from the opposite perspective, if it can be shown
183 that Japanese speakers delete high vowels in accordance with syllable contact law, it may
184 imply that speakers of Japanese, generally considered to be a “CV-language” disallow-
185 ing hetero-organic consonant clusters, are sensitive to wellformedness conditions on con-
186 sonantal clusters (see Berent et al. 2007, 2008 for related ideas).

187 There are other reasons to entertain the current hypothesis. Previous studies have
188 shown that devoicing (not deletion) is more likely between a fricative and a stop than
189 between two fricatives (see e.g. Fujimoto 2015; Hirayama 2009; Maekawa & Kikuchi
190 2005; Martin et al. 2014; Tsuchida 1997). Let us suppose that devoicing and deletion are
191 on the same “reduction continuum”.⁴ Then, everything else being equal, we may expect
192 deletion to be more likely in the environment where devoicing is more likely in the first
193 place. This leads us to expect that deletion is more likely between a fricative and a stop,

⁴In Kagoshima Japanese, word-final high vowels—those that are devoiced—undergo phonological dele-
tion, which feeds other phonological changes of consonants in the final syllables (Haraguchi, 1984; Kaneko
& Kawahara, 2002; Kibe, 2001). It seems natural to consider deletion as the extreme end of the reduction
continuum, and that devoicing is one step in the continuum before deletion (see Haraguchi 1984; McCarthy
2008; Tsuchida 1997).

194 because devoicing is more likely in this environment.

195 A recent acoustic study by Whang et al. (2020) suggests that devoicing and deletion
196 may both be characterized as enhancement strategies of the larynx abduction gesture. Fu-
197 jimoto et al. (2002) as well as Sawashima (1971) show that devoiced vowels in Japanese
198 involve an active abduction gesture, and thus there is a sense in which speakers are ac-
199 tively signalling “voicelessness.” According to Whang et al. (2020), vowel devoicing in
200 fact raises COG of the aperiodic energy of surrounding obstruents, and deletion of the
201 tongue dorsum raising gesture further raises that COG. This hypothesis too leads us to
202 expect that devoicing and deletion should work in tandem with each other, as deletion can
203 enhance the auditory cue to devoicing. To the degree that devoicing is more likely after a
204 fricative and before a stop than between two fricatives (see above), deletion may show the
205 same probabilistic pattern.

206 All of these considerations—prosodic wellformedness, reduction, enhancement of devoicing—
207 converge on the same prediction: deletion should be more likely when it results in a
208 fricative-stop sequence than when it results in a fricative-fricative sequence. Existing ev-
209 idence from Shaw & Kawahara (2018b) is consistent with this conclusion; however, the
210 evidence to date is rather thin.

211 To recap, the current experiment was set out to examine the general hypothesis that
212 vowel deletion probability is modulated by surrounding consonants. The more specific
213 hypothesis is that deletion is most likely between a fricative and a stop, and less likely be-
214 tween two fricatives. The experiment also serves as an attempt to replicate the basic find-
215 ings of Shaw & Kawahara (2018b)—devoiced [u̥]s in Japanese are optionally deleted—
216 with a much extended set of stimuli.

217 **2 Experimental methods**

218 The current experiment measured and analyzed the tongue dorsum trajectories of devoiced
219 [u̥], using EMA (Electro Magnetic Articulatograph). Most of the experimental details fol-
220 low those of Shaw & Kawahara (2018b). One distinct characteristic of this approach that
221 we would like to highlight at this stage is that it assesses the presence of an articulatory
222 target on a token-by-token basis, rather than analyzing averaged contours. This strategy is

important because analyzing averaged contours cannot distinguish two different phonological hypotheses: reduction vs. optional deletion (Cohn, 2006; Kawahara et al., to appear; Shaw & Kawahara, 2018a). Lingual gestures of devoiced vowels, even when not phonologically deleted, are conceivably reduced in magnitude, since the vowel gestures are not as audible due to devoicing and do not contribute much to lexical access anyway (e.g. Hall et al. 2018; Jaeger & Buz 2018; see also Lindblom 1990). Interpreting any difference between voiced vowels and devoiced vowels as deletion would therefore be hasty.

On the other hand, as Shaw & Kawahara (2018b) found, devoiced vowels can retain their full lingual gestures, showing comparable movement trajectories to voiced vowels, but they can also sometimes be deleted. Averaging over cases of full targets and cases of categorical deletion can lead to an erroneous conclusion that the overall pattern supports the reduction hypothesis (Cohn, 2006).

This specific problem can be illustrated by a comparison of two recent studies. Kawahara et al. (to appear) developed a token-by-token analysis of the F0 patterns of the dataset recorded and analyzed by Ishihara (2011). The averaged-based analysis by the latter concluded that pitch accent after wh-elements in Japanese is reduced. On the other hand, a token-by-token reanalysis by Kawahara et al. (to appear) shows that at least some speakers show a mixture of full target and deletion. This comparison shows that when both deletion and reduction are theoretically-justifiable hypotheses, it is important that we distinguish between them through a token-by-token analysis.

In addition to avoiding this general problem of resorting to average-based analyses, the current analysis has a virtue of analyzing the entire articulatory trajectories without relying on “magic moments”; in the current analysis, no aspects of speech signals are given special status, eschewing the potential danger of missing important aspects of dynamic speech (Cho, 2016; Mücke et al., 2014; Vatikiotis-Bateson et al., 2014).

2.1 Participants

Seven native speakers of Tokyo Japanese (4 male) participated in the current experiment. They were all born in Tokyo, lived there at the time of their participation in the study, and had spent no more than 3 months outside of the Tokyo region. Procedures were explained

252 to participants in Japanese by a research assistant, who was also a native speaker of Tokyo
253 Japanese. All participants were naive to the purpose of the experiment. Participants were
254 compensated for their time and local travel expenses. Data from one speaker had to be
255 excluded, because we were unable to record as many repetitions as other speakers. This
256 speaker was originally coded as Speaker 6; their data is not discussed further below. No
257 speakers who participated in Shaw & Kawahara (2018b) participated in this study, since
258 one of the aims was to examine whether the results of Shaw & Kawahara (2018b) can be
259 generalized to other speakers.

260 2.2 Stimuli

261 Following Shaw & Kawahara (2018b), the major target of our analysis is tongue dorsum
262 height in the trajectory of $V_1C_1V_2C_2V_3$ sequence, in which V_2 represents the devoiced
263 vowels in question—justification of this analytical choice is offered below in §3.2, and we
264 will offer some analysis that examines the possibility of whole-sale deletion of devoiced
265 vowels in §6, in addition to the analysis of tongue dorsum height movement. The primary
266 question is whether we would observe a clear rise in tongue dorsum height from V_1 to
267 V_2 and a fall from V_2 to V_3 . V_3 was therefore always a non-high vowel in our stimuli.
268 The target vowels ($V_2=[u]s$) were always word-initial, and V_1 was the last vowel of the
269 preceding word in the frame sentence, [e].

270 At the time of stimulus design, four conditions were included in order to thoroughly
271 explore the effects of surrounding consonant types: fricative-stop (FS), fricative-fricative
272 (FF), stop-stop (SS), and stop-fricative (SF), consisting of 18 dyads shown in Table 1. All
273 the stimuli were existing words in Japanese, where the members on the left were expected
274 to undergo devoicing. Each dyad constituted near minimal pairs, in which one member
275 contained C_1VC_2 sequence where both consonants are voiceless and the other member
276 contained a minimally different C_1VC_2 sequence in which C_2 is voiced, hence V is not
277 expected to devoice.

Table 1: The list of stimuli recorded in the EMA experiment. S=Stop; F=Fricative. See footnote 6 for glosses.

FS	FF
/ϕuton/ vs. /ϕudou/	/ϕusoku/ vs. /ϕuzoku/
/ϕutan/ vs. /ϕudan/	/ϕusai/ vs. /ϕuzai/
/ϕuta/ vs. /ϕuda/	/ϕusagaru/ vs. /ϕuzakeru/
/ʃutaisei/ vs. /ʃudaika/	/ʃusai/ vs. /ʃuzai/
/ʃutou/ vs. /ʃudou/	/ʃusa/ vs. /ʃuzan/
/ʃutokou/ vs. /ʃudouken/	/ʃuso/ vs. /ʃuzou/
SS	SF
/kutakuta/ vs. /kudaranu/	/kusami/ vs. /kuzai/
/kutabaru/ vs. /kudasaru/	/kusari/ vs. /kuzawa/
/kutanijaki/ vs. /kudanʃita/	/kusakari/ vs. /kuzakitʃo/

Choosing existing words with the appropriate segmental compositions did not enable us to control for accentedness within each pair. For example, /ϕutan/ is unaccented, whereas /ϕudan/ is accented on the initial syllable. However, Tsuchida (1997) and Martin et al. (2014) show that accent placement has little effect on devoicing patterns among contemporary speakers of Japanese. Durational differences between accented and unaccented syllables are minimal in Japanese (Beckman, 1986), which if substantial, may affect devoicability/deletability. For these reasons, we judged this difference to be non-crucial.⁵

The current study focused on [u] instead of examining both [u] and [i], both of which are known to devoice. This is partly because the current study is a direct follow-up of Shaw & Kawahara (2018b), who also examined only [u], and also because we needed enough repetitions to execute the computational analysis that was planned (see §3.2 for details). Examining the lingual gesture of devoiced [i] warrants a new set of studies.

After the recording, we came to the conclusion that the conditions in which C_1 is a stop

⁵Shaw & Kawahara (2018b) did not perfectly control for accent between two members within a dyad either, although in their design, [u] is either accented or unaccented within each dyad.

291 (=the SS and SF conditions) could not be reliably analyzed for the following reason. At
292 the time of stimulus design, we decided that C_1 had to be [k], because [p] is not allowed in
293 the native vocabulary (Ito & Mester, 1995), and [t] is affricated before high vowels (Vance,
294 2008). However, since we were interested in the tongue dorsum height of (devoiced) [u],
295 it was not possible to objectively discern control of tongue dorsum height associated with
296 [k] from tongue dorsum height associated with [u]. For this reasons, this paper focuses on
297 the comparison between FS condition and the FF condition.⁶

298 2.3 Procedure

299 Each participant produced 14-15 repetitions of the 36 target words in the carrier phrase:
300 “okkee X to itte” (Ok, say X), where X is a stimulus word. Participants were instructed to
301 speak as if they were making a request of a friend. This was to ensure that the speakers did
302 not speak too formally or too slowly, which may inhibit vowel devoicing in the first place.

303 This resulted in a corpus of 3,204 tokens (14 or 15 repetitions $\times$ 36 words $\times$ 6 speak-
304 ers). Words were presented in Japanese script (composed of hiragana, katakana and kanji
305 characters as required for natural presentation) and fully randomized.

306 2.4 Equipment

307 We used an NDI Wave ElectroMagnetic Articulograph system sampling at 100 Hz to cap-
308 ture articulatory movement. NDI wave 5DoF sensors (receiver coils) were attached to
309 three locations on the sagittal midline of the tongue, and on the lips, jaw (below the lower
310 incisor), nasion and left/right mastoids. The most anterior sensor on the tongue, henceforth
311 TT, was attached less than one cm from the tongue tip (see Figure 1). The most posterior
312 sensor, henceforth TD, was attached as far back as was comfortable for the participant. A
313 third sensor, henceforth TB, was placed on the tongue body roughly equidistant between
314 the TT and TD sensors. Sensors were attached with attached with a combination of sur-

⁶The glosses for the items that were analyzed are as follows. FF: blanket vs. not moving, burden vs. usual, top vs. amulet, subjectivity vs. main theme, FOOD NAME vs. hand-moving, Tokyo Highway vs. lead; FS: shortage vs. attachment, debt vs. absence, filled vs. joke, organize vs. research, chair vs. abacus, main claim vs. *sake*-making.

315 gical glue and ketac dental adhesive. Acoustic data were recorded simultaneously at 22
316 KHz with a Schoeps MK 41S supercardioid microphone (with Schoeps CMC 6 Ug power
317 module).

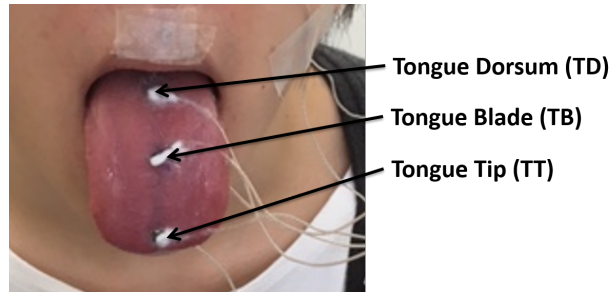


Figure 1: Illustration of the sensor placement (reproduced from Shaw & Kawahara 2018b).

318 **2.5 Stimulus display**

319 Words were displayed on a monitor positioned 25cm outside of the NDI Wave magnetic
320 field. Stimulus display was controlled manually using an Eprime script. This setup allowed
321 for online monitoring of hesitations, mispronunciations and disfluencies. These were rare,
322 but when they occurred, items were marked for repeated presentation by the experimenter.
323 These items were then re-inserted into the random presentation of remaining items. This
324 method ensured that we recorded at least 14 fluent tokens of each target item.

325 **2.6 Post-processing**

326 Following the main recording session, we also recorded the bite plane of each participant
327 by having them hold a rigid object, with three 5DoF sensors attached to it, between their
328 teeth. Head movements were corrected computationally after data collection with refer-
329 ence to three sensors on the head, the left/right mastoid and nasion sensors, and the three
330 sensors on the bite plane. The head corrected data was rotated so that the origin of the
331 spatial coordinates corresponds to the occlusal plane at the front teeth.

3 Data analysis

3.1 Data processing

The wav files recorded in the experiment were submitted to forced alignment, using FAVE.⁷ Textgrids from forced alignment were hand-corrected and, during this process, the target vowels were coded for voicing. Many vowels in devoicing environments were in fact devoiced, as evident from visual inspection of the spectrogram and waveform. However, some tokens in the devoicing environment retained clear signs of glottal vibration. These vowels were coded as voiced, and excluded from the following computational analysis.

Articulatory data corresponding to each token were extracted based on the textgrids. To eliminate high frequency noise in the recording, the data were smoothed using the robust smoothing algorithm (Garcia, 2010) and, subsequently, visualized in MVIEW, a Matlab-based program to analyze articulatory data (Tiede, 2005). Within MVIEW, the consonant gestures flanking the target vowel were parsed using the `findgest` algorithm. `Findgest` identifies gestures semi-automatically based upon the velocity signal in the movement toward and away from gestural targets. An illustrative example is provided in Figure 2. The consonant gestures were used to define a temporal interval for further analysis. Tokens with missing data in the target interval were excluded from further analysis. Some tokens had velocity peaks that were not large enough to clearly parse out movement related to the consonants. If a token was missing a gesture parse for either consonant, it was excluded from further analysis. A total of 239 tokens were excluded for this reason. The resulting data set consisted of 2,431 tokens for analysis, which had clearly distinguishable consonantal gestures flanking the target vowel.

⁷<https://github.com/JoFrhwld/FAVE/wiki/Using-FAVE-align>

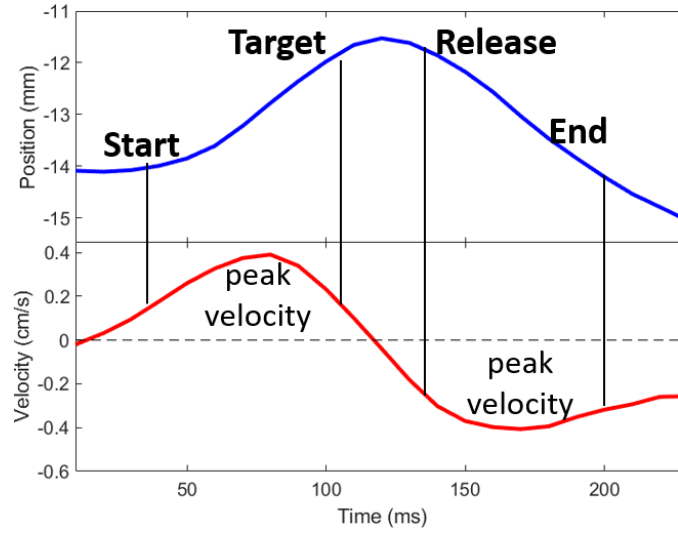


Figure 2: A sample articulatory trajectory and how the articulatory landmarks were identified using `findgest`.

3.2 Computational analyses

The temporal interval spanning from the onset of movement of C_1 , the consonant preceding the target vowel, and the offset of movement of C_2 , the consonant following the target vowel, was subjected to further analysis. To address the question of whether devoiced [u] has an articulatory target, we focused on tongue height, instead of tongue retraction or lip gestures, both of which have been questioned as reliable articulatory correlates of this vowel in contemporary Japanese (Isomura, 2009; Nogita et al., 2013; Shaw & Kawahara, 2018a; Vance, 2008). Like Shaw & Kawahara (2018b), the analysis focused on the movements of the TD sensor (see Figure 1), the most posterior sensor on the tongue, which is typically used to detect vowel gestures (Browman & Goldstein, 1992b; Johnson et al., 1993). We will come back to the issue of whether entire vowels—not just tongue dorsum gesture—are deleted in §6.

Figure 3 shows sample trajectories of a voiced vowel (left), a devoiced vowel with a clear tongue dorsum raising during [u] (middle), and a devoiced vowel without a very

368 clear movement in terms of tongue dorsum height (right). The top panels show the audio
 369 signal. The second panels from the top show tongue dorsum articulatory trajectories,
 370 which are the primary target of our analyses. For reference the third and fourth panels
 371 show trajectories related to the flanking consonants. The token in the right panel does not
 372 appear to have a clear tongue dorsum raising gesture during [u], whereas the [u] token
 373 in the middle panel does seem to have a clear raising gesture. The challenge is to go
 374 beyond such impressionistic classifications and to establish an objective method to classify
 375 whether devoiced vowels show a tongue dorsum raising gesture or not.

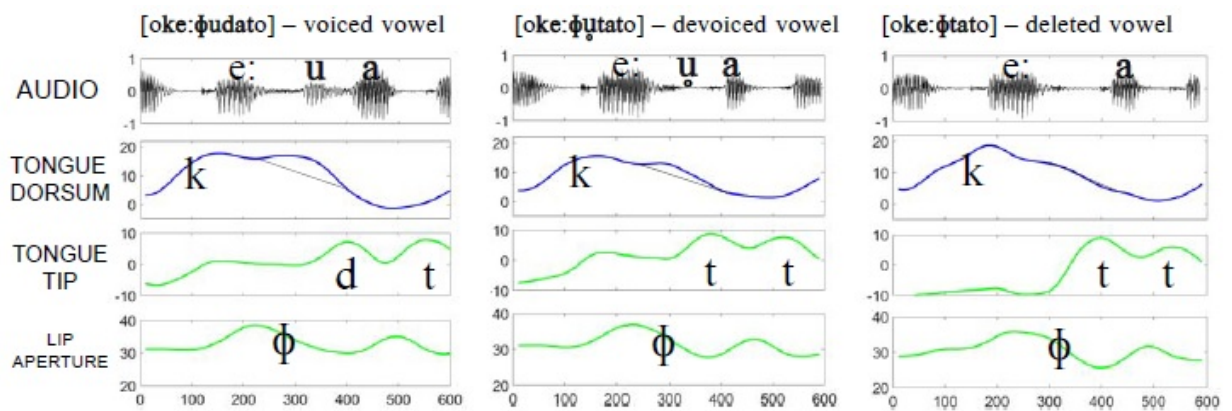


Figure 3: Sample EMA trajectories. The top panels show audio signals. The second panels show the tongue dorsum movement. A linear interpolation line from the preceding vowel to the following vowel is superimposed. This line is not very visible in the right panel, because this token does not show a clear tongue dorsum rise.

376 To do so, we applied the approach described and motivated in detail in Shaw & Kawa-
 377 hara (2018a,b), schematically illustrated in Figure 4. This computational methodology
 378 was developed to assess the presence/absence of a lingual vowel target of devoiced vowels
 379 in articulatory trajectories. The approach is general enough that it has been extended to
 380 other types of continuous phonetic data, including nasal reduction in Ende (Brickhouse &
 381 Lindsey, 2020), pitch accent eradication in Japanese (Kawahara et al., to appear), and tone
 382 reduction in Mandarin Chinese (Zhang et al., 2019).

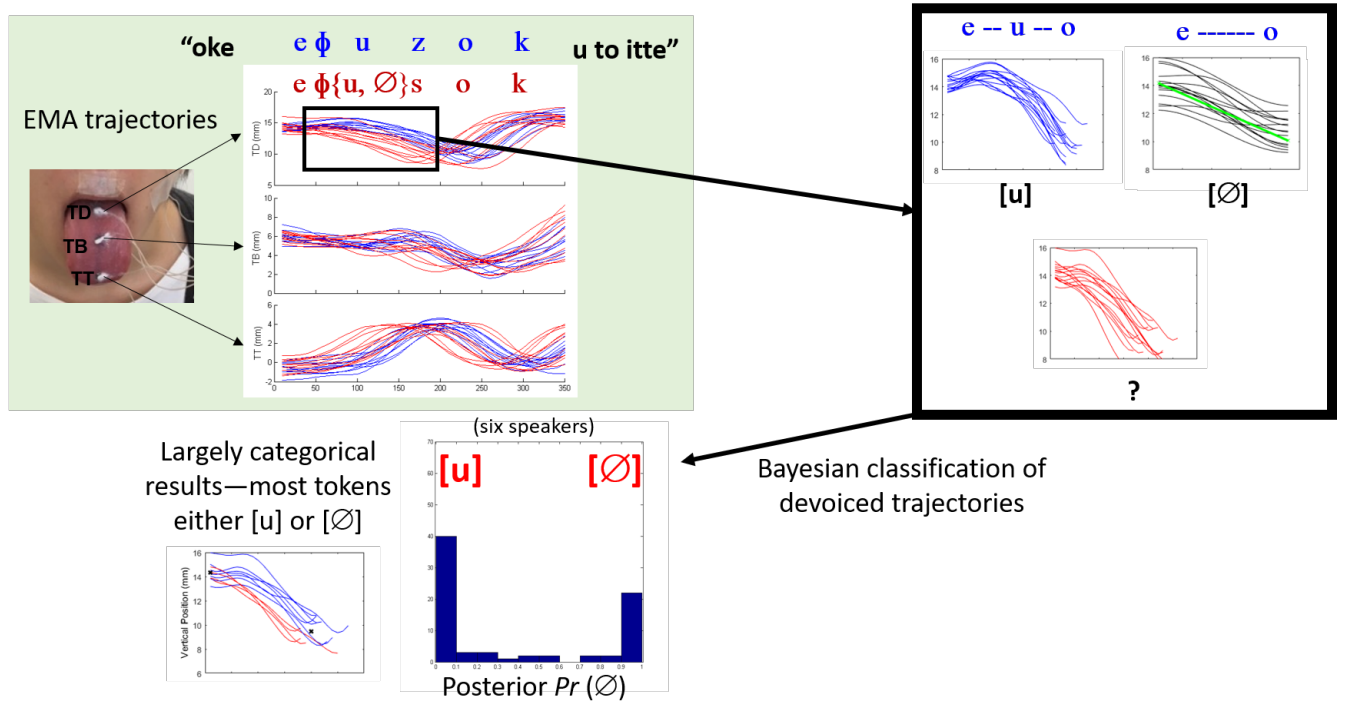


Figure 4: Summary of simulation and classification procedure developed and defended in Shaw & Kawahara (2018b).

383 The target interval spans from the preceding vowel to the following vowel (see the left
 384 upper panel of Figure 4). For example, for the word [φuzoku], the analysis window starts
 385 from [e] in the carrier sentence and includes the main target CVC ([φuz]) and the following
 386 vowel [o]. The question of interest is whether given the vowel sequence [e]-[u]-[o], we
 387 would observe a tongue dorsum raising gesture, when [u] is devoiced. When [u]’s tongue
 388 dorsum gesture is undoubtedly present, as in the case for voiced [u], we should observe a
 389 clear raising gesture (the left panel of Figure 3). On the other hand, if the vowel gesture
 390 is deleted, we observe articulatory trajectories that are best viewed as an interpolation
 391 between [e] and [o] (represented as a green straight line in the right upper panel of Figure
 392 4). Since articulatory movements, as behavioral data more generally, are always noisy
 393 actuations of intentions, the challenge is to develop an objective method with which we

394 can assess whether each articulatory contour of a devoiced [u] is better characterized as
395 target-present or target-absent (the upper right box in Figure 4). The computational toolkit
396 developed by Shaw & Kawahara (2018a,b) allows us to address this question on a token-
397 by-token basis.

398 The first step in this computational method is to analyze the articulatory trajectories
399 in a low-dimensional space, by making use of Discrete Cosine Transform (DCT) (e.g.
400 Jain 1989). Through DCT, a signal is transformed into the sum of cosine components of
401 gradually increasing frequency. This transformation is similar to Fourier transform in that
402 timeseries data—here, the articulatory trajectory—is represented in frequency space, i.e.,
403 as cosines of varying frequency and magnitude. Unlike Fourier transform, DCT uses only
404 cosines instead of a combination of sines and cosines and there is no imaginary component.
405 For our purposes, DCT is useful because of its compression properties (Jain, 1989)—
406 the articulatory trajectory within the analysis window can often be represented with a
407 small number of DCT components. Because speech articulators are relatively slow, high
408 frequency components are not needed to represent their controlled movement, a point
409 which we demonstrate below.

410 The numerical expression of DCT is provided in Equations (1) and (2): n is the po-
411 sitional signal, L is the length of the window (in samples), k is the number of the DCT
412 coefficient, which ranges from 1 to L , y is the magnitude of each coefficient, and w is a
413 weight. DCT coefficients can be positive or negative and their absolute value represents
414 the magnitude of their contribution to spatial modulation of the signal. For the first DCT
415 coefficient, the numerator in the scope of the cosine is zero, which means that it equals
416 1 for every sample n in the trajectory. These are summed, and when multiplied by the
417 relevant weight ($\frac{1}{\sqrt{L}}$), they yield a quantity that is related to the average of the trajectory
418 (if the weight was $\frac{1}{L}$, then it would be the average). This first cosine coefficient serves as a
419 baseline, c.f. the intercept in a linear regression. As k increases beyond one, the resulting
420 cosines gradually increase in frequency, $k = 2$ yields a cosine that completes one quarter
421 of its cycle within the signal, $k = 3$, yields a half cycle and so on (see Figure 6). DCT
422 produces $k = L$ components, so the number of cosine components depends on the length
423 of the signal. However, the magnitude of the higher frequency components may be quite
424 small for signals of slow moving articulators.

$$y(k) = w(k) \sum_{n=1}^L \cos \frac{\pi(2n-1)(k-1)}{2L} \quad k = 1, 2, \dots, L \quad (1)$$

where

$$w(k) = \begin{cases} \frac{1}{\sqrt{L}} & k = 1 \\ \sqrt{\frac{2}{L}} & 2 \leq k \leq L \end{cases} \quad (2)$$

425 DCT has a known inverse function, iDCT, which can be used to simulate trajectories
426 from DCT components (= Equations (3) and (4)).

$$x(n) = \sum_{k=1}^L w(k)y(k) \cos \frac{\pi(2n-1)(k-1)}{2L} \quad n = 1, 2, \dots, L \quad (3)$$

where

$$\begin{cases} \frac{1}{\sqrt{L}} & k = 1 \\ \sqrt{\frac{2}{L}} & 2 \leq k \leq L \end{cases} \quad (4)$$

427 We make use of iDCT to assess how many DCT components are necessary to faith-
428 fully represent the actual articulatory trajectories. We do this by fitting DCT components
429 to a set of trajectories and then resynthesizing using iDCT with progressively more DCT
430 components. In this way, we can observe how increasing the number of DCT components
431 improves the precision of the representation. Figure 5 shows representative results, from
432 one speaker and one item ([futokou] produced by Speaker 7). The improvement from 1
433 DCT component to 2 is substantial, as is the improvement from the 2 components to 3
434 components. With four components the correlation between the raw trajectories and the
435 iDCT-simulated trajectories reaches $r = 0.99$. In our case, only a small number of DCT

436 components (3 or 4) are required to faithfully represent articulatory trajectories over the
 437 target VCVCV window. This result is similar to past studies, which have modelled tra-
 438 jectories of similar duration and linguistic complexity using either 3 (Shaw & Kawahara,
 439 2018a) or 4 (Shaw & Kawahara, 2018b; Kawahara et al., to appear) components.

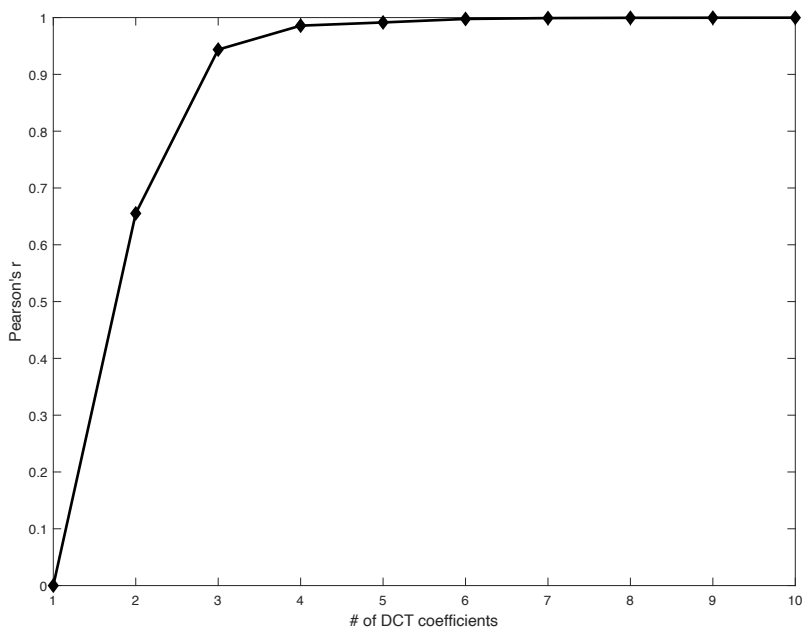


Figure 5: The increase in Pearson coefficients between the number of DCT components and the correlation between actual trajectories and simulated trajectories.

440 We can also use iDCT to illustrate how each DCT component contributes to the rep-
 441 resentation of the articulatory trajectory. The top panel of Figure 6 shows the average
 442 articulatory trajectories for each item of the dyad, [futokoo] (left) vs. [judookoo] (right).

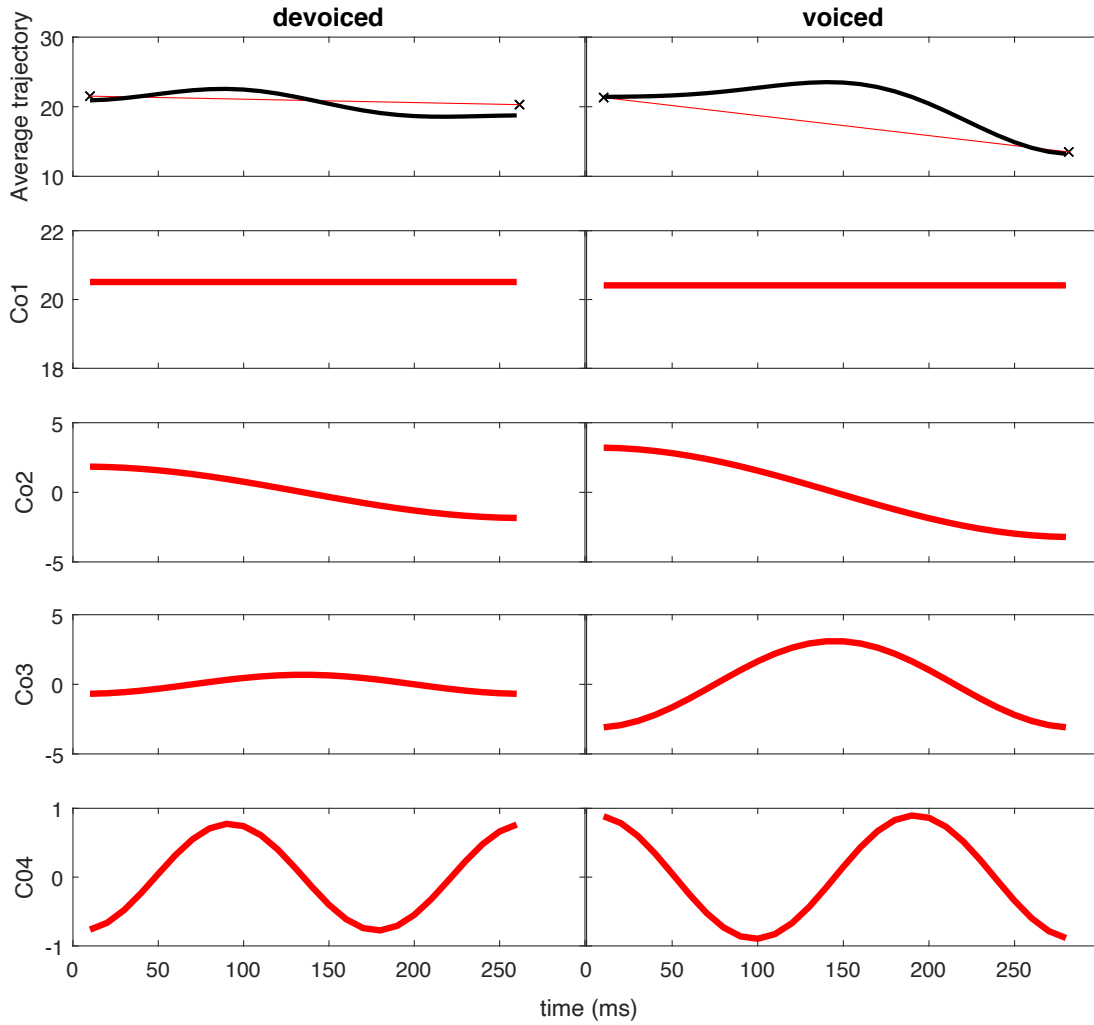


Figure 6: A sample comparison between the four DCT components of articulatory trajectories of devoiced and voiced tokens (averaged).

443 Given this dyad, we can observe that the average change in tongue dorsum height over
 444 time, shown in the top panel, is noticeably different between devoiced and voiced items.
 445 For the voiced item (right), the tongue dorsum rises in the middle of the trajectory for [u].
 446 For the devoiced item, there is less variation in the positional signal over time. For refer-

447 ence, the “x”s in the top panel show the average position at the start and end of the analysis
448 window. The straight line connecting the x-points is equivalent to a linear interpolation of
449 spatial position across the analysis window. The panels below the trajectory show the con-
450 tribution of each DCT component to spatial modulation of the signal. Comparison across
451 devoiced and voiced items reveals similar modulations for the first coefficient (Co1) and
452 the second coefficient (Co2). The main difference is in the third (Co3) and fourth (Co4)
453 coefficients. Co3 picks up on the large rise for [u] in the voiced case.⁸ The magnitude of
454 the rise contributed by Co3 is greatly reduced for the devoiced item compared to the voiced
455 item. Finally, the fourth DCT coefficient (Co4) is also quite different between voiced and
456 devoiced items but it has only a small effect on spatial position overall—note that the scale
457 on the y-axis for Co4 ranges from 1 to -1 (units are in mm) instead of 5 to -5, like Co2 and
458 Co3.

459 The next step is to assess whether the devoiced item contains a vowel target or not.
460 To do this we set up stochastic generators of our competing hypotheses, which we use for
461 Bayesian classification. The “target present” hypothesis is based on the voiced member
462 of each dyad. Specifically, since we have multiple repetitions of each item, we can cal-
463 culate a distribution over each DCT component. The normal distribution is characterized
464 by a mean value and a standard deviation. Thus, the mean and standard deviation of each
465 DCT component characterizes a normal probability distribution function. For the “target
466 absent” case, we adopt the common assumption that, in the absence of phonological spec-
467 ification, the trajectory will interpolate between surrounding targets (Choi, 1995; Cohn,
468 1993; Keating, 1988; Pierrehumbert & Beckman, 1988). We therefore construct prob-
469 ability distributions for the “target absent” hypothesis that capture a realistically noisy
470 interpolation. For each token of a devoiced item, we fit DCT components to the straight
471 line connecting the position at the onset and offset of the analysis window.⁹ The average of
472 these components defines the probability distributions for the “target absent” hypothesis.
473 The standard deviation for the distributions is computed from the devoiced trajectories in
474 the same manner as for the voiced item. Consequently, the probability distributions that

⁸We note however that it is not necessarily the case that each DCT coefficient has to have a meaningful linguistic interpretation; neither is it the case that we have reasons to believe that Co3 is solely responsible for representing the tongue dorsum raising gesture of [u].

⁹See Pierrehumbert (1980) and Myers (1998) for cases of non-linear interpolation.

characterize the “target absent” hypothesis are defined by linear interpolation (means of the distribution) and the variability around each DCT component in the data. An example of the resulting distributions is provided in Figure 7. The horizontal axis is the value of the coefficient, i.e., y in Equation (1), and the vertical axis is probability.

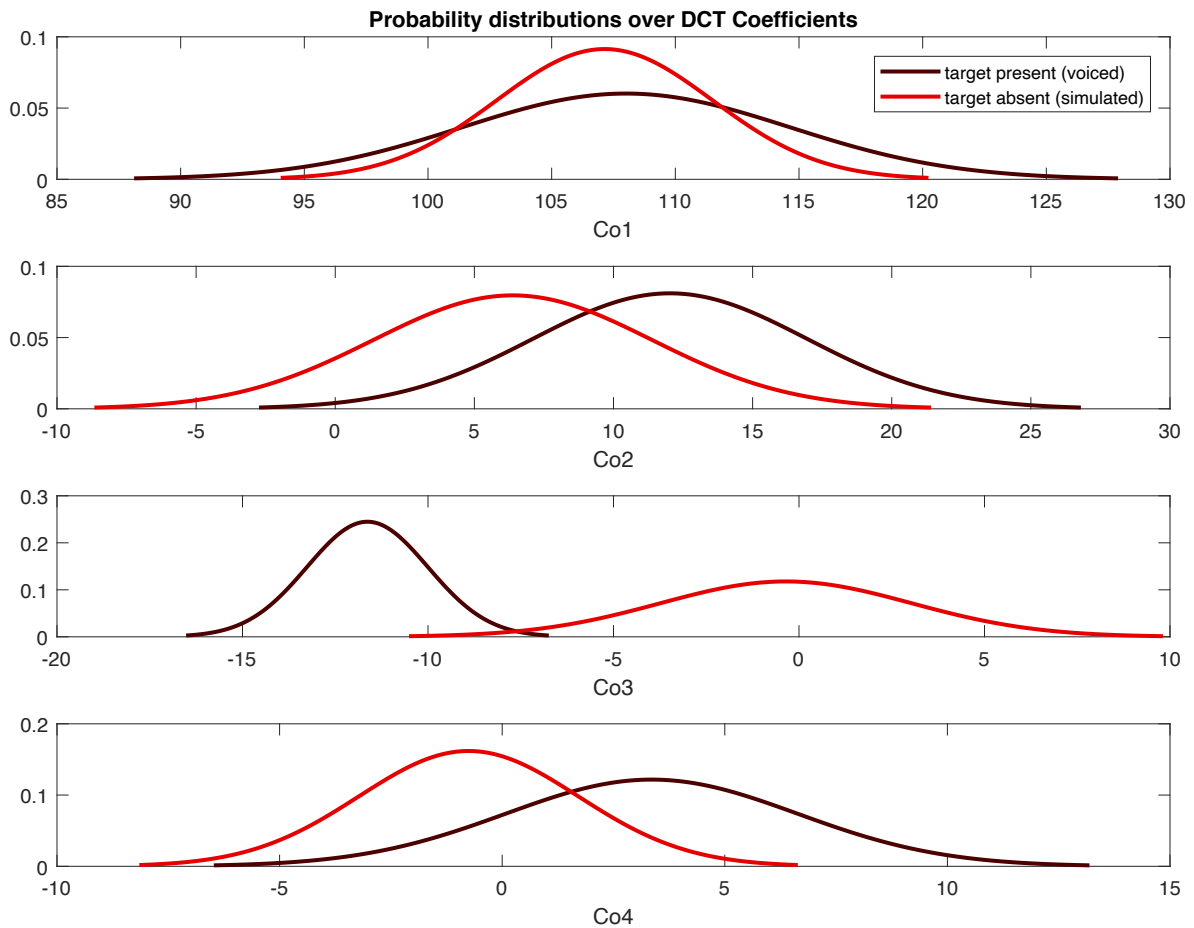


Figure 7: Probability distributions for DCT coefficients for the two competing hypotheses. The “target present” condition is based on the voiced vowels. The “target absent” condition is based on linear interpolation and the level of variability in the devoiced vowels.

We observe that the distributions for Co1 between the two conditions overlap heavily.

480 For Co2, there is a small difference between the “target present” distributions, based on
 481 voiced vowels, and “target absent” distribution, based on linear interpolation. The largest
 482 difference appears to lie in Co3. Naturally, the mean of the “target absent” distribution
 483 is very close to zero, and the same goes for Co4. This is because there is no rise for the
 484 straight line fit connecting the positional signal at the onset and offset of the analysis win-
 485 dow. The “target absent” Co3 distribution is also more variable than the corresponding
 486 “target absent” distribution—this difference reflects greater variability across devoiced to-
 487 kens than voiced tokens in whether the trajectory showed a rise characteristic of a vowel
 488 or not.

489 As the final step of the computational analysis, for each devoiced token, we determined
 490 the posterior probability of a vowel height target, based on Bayesian classification of the
 491 tongue dorsum trajectory (=Equation (5)). The posterior probability of the targetless hy-
 492 pothesis given the set of DCT coefficients (the left term of the Equation) is expressed as the
 493 prior probability of the targetless hypothesis—always set to be 0.5 in the current analysis—
 494 multiplied by the product of the conditional probabilities of each DCT coefficient given
 495 the targetless hypothesis (i.e. linear interpolation), normalized by the denominator term.
 496 The classifier was trained on the distributions described above (see Figure 7) for voiced
 497 tokens, which unambiguously contain a vowel target, and a noisy null hypothesis, defined
 498 as linear interpolation across the target interval.

$$p(T|Co_1, \dots, Co_n) = \frac{p(T) \times \prod_{i=1}^n p(Co_i|T)}{\prod_{i=1}^n p(Co_i)} \quad (5)$$

499 To summarize, the approach described in this subsection assigns a probability of target
 500 absence to each token. It does so by considering the probability that the token follows a
 501 linear interpolation as opposed to the trajectory of voiced vowels.

502 4 Results

503 Figure 8 shows the posterior probability of target absence for each condition by each
 504 speaker. The figures are violin plots which show the distribution of posterior probabilities
 505 of target absence. Points around the high y-axis region are tokens with a high probability

506 of target absence, i.e., lingual movements that can be characterized as linear interpolation
 507 through the devoiced portion of the signal. Those at the bottom of the y-axis are tokens
 508 that have a high probability of a vowel target, i.e., lingual articulations that resemble the
 509 voiced tokens. Those in the middle range are intermediate between target present and
 510 target absent, indicating a spatially reduced vowel target.

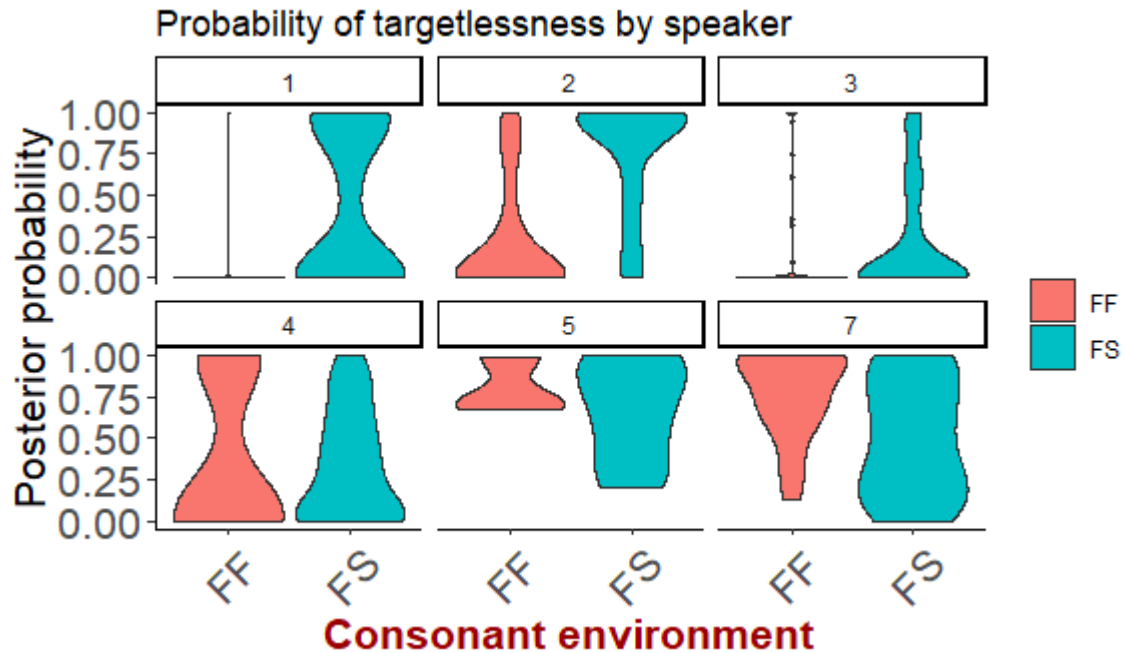


Figure 8: Posterior “target absent” probability for each condition by speaker. FF=Fricative-Fricative; FS=Fricative-Stop.

511 We observe that, as with Shaw & Kawahara (2018b), most data points distribute at low
 512 or high ends of the y-axis, although we see some tokens that are intermediate (e.g. the
 513 FF condition for Speakers 5 and 7). Overall, therefore, we can conclude that in general,
 514 devoiced vowels in Japanese tend to either fully retain their lingual gesture or entirely
 515 lose it. This result replicates the findings by Shaw & Kawahara (2018b) with a new set
 516 of speakers and an expanded set of stimuli. Recall that the study by Shaw & Kawahara
 517 (2018b) examined only four dyads; the current results are based on twelve dyads.

518 How the flanking consonants influenced targetless probability varied between speak-
519 ers. Speaker 1 showed almost no targetless tokens in the FF condition, but showed some
520 targetless tokens in the FS condition. This pattern—more targetlessness in the FS condi-
521 tion than in the FF condition—accords well with the prediction laid out in § 1.3. Speaker 2
522 shows a similar, and perhaps clearer, pattern; this speaker showed rather consistent target-
523 present production in the FF condition, but typically deleted tongue dorsum raising gesture
524 in the FS condition. The pattern exhibited by Speaker 3 is less clear, but is also consistent
525 with the hypothesis presented in § 1.3: almost no targetless tokens in the FF condition,
526 but greater probability of targetlessness in the FS condition. These three speakers thus
527 confirmed the hypothesis that we formulated in § 1.3.

528 However, not all speakers behaved as we hypothesized. Speaker 5, especially in the
529 FF condition, seems to show some tokens whose posterior probabilities are in the middle
530 range—those tokens that are neither clearly targetless nor have a full target. Speakers 4 and
531 7, especially the latter, showed a pattern that is opposite from what is predicted from the
532 considerations discussed in § 1.3—more targetless tokens in the FF condition than in the
533 FS condition. Thus, looking across the six speakers, we observe speaker-specific variation
534 in whether FF or FS environments conditions more deletion of the tongue dorsum raising
535 gesture.

536 Figure 9 shows the results by item. From this plot we can see some variability across
537 items as well. For example, / ϕ usagaru/, the only verb in the item list, shows the lowest
538 probability of targetlessness. Many words show fairly sharp bi-modal patterns, with some
539 tokens showing high probability of targetlessness and others showing high probability of
540 full targets with few intermediate tokens. This bi-modal pattern applies especially clearly
541 to / ϕ uta/, / ϕ utan/, / ϕ uton/, /fusa/, and /futokou/. In contrast, most tokens of /fusai/ are
542 intermediate, with few extreme probabilities in either direction.

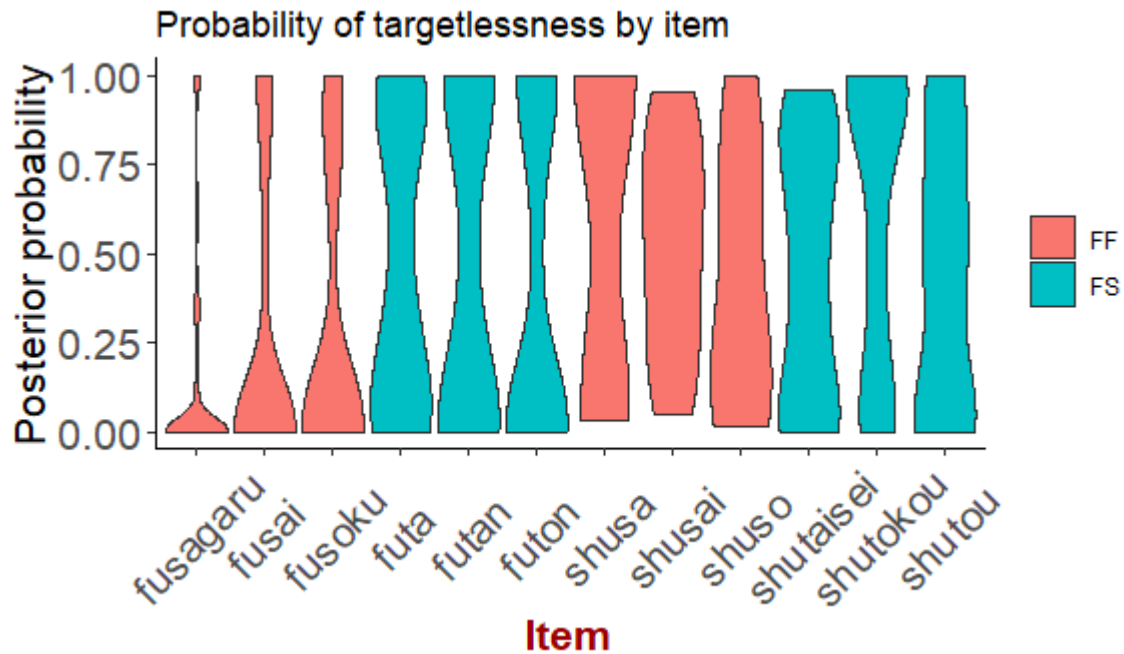


Figure 9: Posterior “target absent” probability by item. “f” and “sh” are used in the figure in place of / ϕ / and / $\jmath$ /, respectively.

543 To assess the overall results statistically, we fit a series of nested linear mixed effects
544 models in (7). The results of model comparisons appear in Table 2. The dependent vari-
545 able was the posterior probability of deletion. Since probabilities are bounded dependent
546 variables (upper bound of 1; lower bound of 0), we also ran the same models on arcsin-
547 transformed probabilities. The same pattern of results came out of both raw and trans-
548 formed probabilities. For reasons of space we report results based on the non-transformed
549 probabilities. The key fixed effect of interest was the consonant environment, coded as a
550 two level factors, FF vs. FS (“*Cond*”). Speakers and items were treated as random inter-
551 cepts.

$$\begin{aligned}
m0 : post &\sim (1|speaker) + (1|item) \\
m1 : post &\sim (1 + Cond|speaker) + (1|item) \\
m2 : post &\sim Cond + (1 + Cond|speaker) + (1|item) \\
m3 : post &\sim C1 + (1 + Cond|speaker) + (1|item) \\
m4 : post &\sim C1 * Cond + (1 + Cond|speaker) + (1|item)
\end{aligned} \tag{6}$$

Table 2: Summary of model comparisons.

	df	AIC	BIC	logLik	deviance	χ^2	χ^2 df	p
$m0$	4	464.7	481.7	-228.3	456.7	—	—	—
$m1$	6	402.6	428.1	-195.3	390.6	66.07	2	< .001
$m2$	7	404.1	433.8	-195.0	390.1	0.53	1	<i>n.s.</i>
$m3$	7	400.4	430.1	-193.2	386.4	4.25	1	< .05
$m4$	9	403.7	441.9	-192.8	385.7	4.95	3	<i>n.s.</i>

552 The baseline model, $m0$, includes only the random effects. The next model, $m1$, adds
553 a by-speaker random slope for the fixed effect, i.e. surrounding consonants (FF vs. FS) to
554 this model. The by-speaker random slope improved the model significantly. This result
555 indicates that speakers show different sensitivities to the consonantal environments. As
556 we observed in Figure 8, some speakers (e.g. Speakers 1 and 2) show less deletion in FF
557 than FS environments, while others (Speakers 4 and 7) show the opposite pattern.

558 Because the effect of consonant environment differs by speaker, the average effect of
559 consonantal environment is not predictive. These statistical comparisons support what
560 we observed in Figure 8: different speakers are sensitive to consonantal environment in
561 different ways.

562 We also ran models that included the C_1 type ([ϕ] vs. [j]) and the interaction between
563 C_1 and consonant environment (“*Cond*”) as fixed factors. The addition of C_1 led to im-
564 provement over $m1$, and was marginally significant within the model ($\beta = 0.098, t =$
565 $2.136, p = 0.055$), indicating that deletion probability is slightly higher when C_1 is / j / than

566 when C_1 is / ϕ /. The interaction between C_1 and consonant environment (“*Cond*”) did not
567 lead to further improvement, indicating that the effect of C_1 is not dependent on the con-
568 sonant sequence. Thus, our best fitting model, m3, includes a (“*Cond*”) as random effect
569 but not as a fixed effect.

570 Figure 10 shows the by-speaker random slopes for our best fitting model. The x-axis
571 shows the estimate for FS sequences. As we observed in the violin plots of probabilities
572 (=Figure 8), Speakers 1 and 2 have positive estimates, indicating that deletion is more
573 likely in FS sequences than in FF sequences. Moreover, the confidence intervals around
574 the estimate do not overlap with zero. Additionally, as we also observed above, Speaker
575 7 shows the opposite pattern. This speaker has a negative estimate, which also does not
576 overlap with zero, indicating significantly higher probability of targetlessness in FF se-
577 quences than in FS sequences. The other speakers have estimates that overlap with zero,
578 indicating an effect that is not statistically significant.

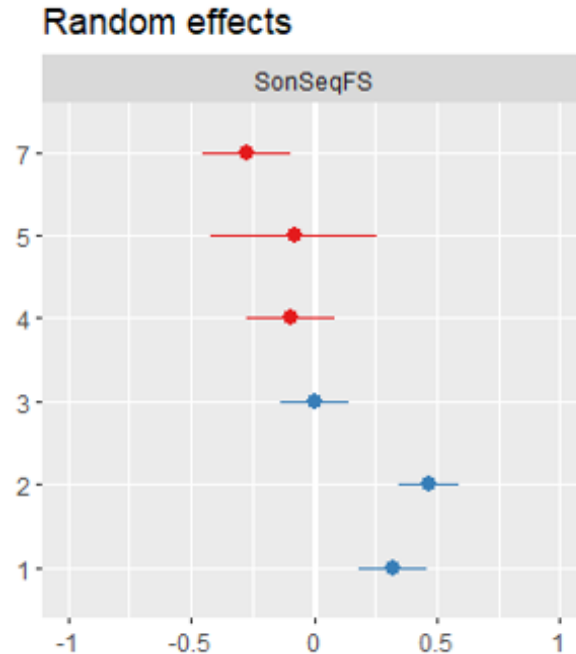


Figure 10: By-speaker random slopes for the effect of sonority sequencing (=Cond). The estimate is for the FS condition, relative to FF.

579 In summary, consonant environment had a significant impact on deletion probability,
 580 but the direction of the effect was not uniform across speakers. Some speakers showed
 581 consistently more deletion in FS, as predicted, others showed more deletion in FF, or no
 582 effect of consonant context.

583 5 Discussion

584 The current experiment replicated the core finding of Shaw & Kawahara (2018b) with a
 585 new set of speakers and an extended set of stimuli: devoiced [u̥]s in Japanese tend to either
 586 fully retain their tongue body gesture or to lack a tongue body gesture entirely. Tokens
 587 with articulatory movements of intermediate magnitude between full vowels (represented
 588 by voiced tokens) and no vowels (represented by interpolation) were rare, and much less

589 common than extremes (see, e.g., Figure 8).

590 The experiment also revealed inter-speaker variability with respect to the prediction
591 laid out in §1.3: those who showed more targetless tokens in the FS condition than in the
592 FF condition (Speakers 1 and 2), as we initially hypothesized, and those who showed the
593 opposite pattern (Speaker 7, and to a less clear extent, Speaker 4). The original hypothesis
594 developed in §1.3 is supported by the behavior of the first group of speakers. Recall that,
595 to the extent that we can conceive of deletion as an extreme instantiation of devoicing,
596 either in terms of reduction or enhancement, we would expect targetless tokens to be more
597 likely in the FS condition than in the FF condition, because devoicing is more likely in
598 this environment. Syllable contact laws (Murray, 1988; Murray & Vennemann, 1983), if
599 Japanese speakers are sensitive to them, also predict this pattern.

600 The behavior of the second group of speakers, on the other hand, is puzzling in light
601 of the hypothesis we presented in §1.3. One admittedly post-hoc speculation is that these
602 speakers may retain a lingual gesture before a stop more often than before a fricative due
603 to a perceptual reason. The perceptual information regarding the place of articulation of
604 stops is at least partly conveyed in the preceding vowel (Fujimura et al., 1978; Steriade,
605 2001; Wright, 1996, 2004), whereas fricatives contain internal cues to place of articula-
606 tion (e.g. Morelli 1999; Wright 1996, 2004 though cf. Li et al. 2009). To the extent that
607 lingual gestures can preserve cues for place of articulation of following stops, Speakers 4
608 and 7 may be leaning toward retaining lingual gestures in this context in order to convey
609 place information of the following stop. If so, the inter-speaker variation that we found
610 in the current experiment could reflect the competing demands of obeying prosodic well-
611 formedness conditions (e.g. syllable contact law) and maximizing the recoverability of
612 the following stop; high vowel devoicing as a sociolinguistic marker of prestige in Tokyo
613 Japanese (Imai, 2004) may also be at play—speakers may feel pressure to enhance devoic-
614 ing to signal their sociolinguistic identity, even at the cost of cues to recovering flanking
615 consonants. That said, whether maintaining the lingual gesture even in the presence of
616 devoicing actually facilitates recoverability of the following stop is a post-hoc hypothesis
617 and remains an open empirical question.

618 **6 Are whole vowels deleted? Analyses of ICI**

619 **6.1 Introduction**

620 The analysis of the experiment reported in §3-§5 exclusively focused on tongue height.
621 There remains a possibility, however, that vowels lacking a height target may nevertheless
622 retain other features, e.g. tongue dorsum retraction, a lip gesture or even an “empty timing
623 slot”, although as we mentioned in §3.2, previous studies suggest neither tongue retraction
624 nor lip rounding as reliable correlates of Japanese [u] (Isomura, 2009; Nogita et al., 2013;
625 Shaw & Kawahara, 2018a; Vance, 2008). Nevertheless, to address the question of whether
626 those vowels that were classified as lacking a height target are phonologically deleted, we
627 next explored patterns of inter-gestural coordination.

628 The basic reasoning, as developed in Shaw & Kawahara (2018b), is as follows . If
629 the vowel is present, then the vowel can be coordinated in time both with the preceding
630 and following consonants. On the other hand, if the vowel is entirely deleted, the two
631 flanking consonants need to be coordinated with each other. For concreteness, consider the
632 coordination topology represented in Figure 11. The top left panel shows a coordination
633 topology of C_1VC_2 sequences. We posit that C_1 and following vowel are timed in-phase
634 (synchronously), while the vowel and C_2 are timed anti-phase (sequentially). However,
635 if the vowel is completely deleted, as in the right panel, then this coordination schema
636 is no longer available. That is, C_2 must be timed immediately to C_1 , as shown in the
637 coordination topology on the right. We posit that these consonants are timed anti-phase,
638 such that C_2 begins when gestural control of C_1 ends.

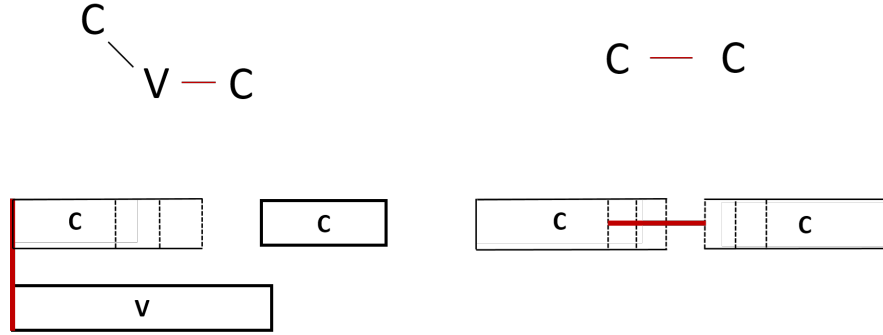


Figure 11: Different coordination topology patterns in C_1VC_2 sequences and C_1C_2 sequences. Adapted from Shaw & Kawahara (2018b).

639 The different coordination topologies in Figure 11 make different predictions about
 640 the covariation between C_1 duration and the inter-consonantal interval (ICI), defined as
 641 the interval from the release of C_1 to the achievement of target of C_2 (Shaw & Kawahara,
 642 2018b). When the vowel is present (the left panel of Figure 11), increases in C_1 duration
 643 will, all else equal, decrease ICI, because ICI is fixed in this coordination pattern. Thus,
 644 there should be a negative correlation between these intervals (C_1 duration and ICI) when
 645 the vowel is present. When the vowel is absent (the right panel of Figure 11), on the
 646 other hand, variation in C_1 duration is not predicted to impact ICI, because the onset of
 647 C_2 is coordinated with the offset of C_1 , i.e. C_2 onset can covary with C_1 offset. Shaw
 648 & Kawahara (2018b) demonstrate that these different covariation patterns indeed hold in
 649 their dataset; we use these predicted patterns of covariation to assess the completeness of
 650 vowel deletion in the current dataset.

651 6.2 Methods

652 We used the same gestural landmarks shown in Figure 2. Consonant duration was defined
 653 as the difference between the release and target landmarks. This provides an estimate of
 654 consonant closure duration similar to acoustic segmentation of consonant duration. The
 655 inter-consonantal interval (ICI) was defined as the interval from the release of C_1 to the

achievement of target of C_2 . To classify tokens as either CC (vowel absent) or CVC (vowel present), we applied the Bayesian decision rule to the posterior probabilities of targetlessness. Those tokens with higher than a .5 probability of lacking a target were classified as belonging to the CC condition. This criterion is typical in the application of Bayesian classifiers, and it is particularly appropriate in our case since the probabilities tended to show a bimodal distribution, with most tokens clustered near either 1 or 0 and few values close to .5. At issue is whether the population of tokens classified as vowel absent based solely on tongue dorsum height show patterns of covariation also characteristic of CC (vowel absent) tokens.

6.3 Results

Figure 12 shows a scatter plot for the tokens classified as lacking a vowel height target, CC (left), and those classified as having a vowel height target, CVC (right). As predicted, for the CVC condition, there is a strong negative correlation between the interconsonantal interval (ICI) and C_1 duration. We can attribute this correlation to the gestural coordination pattern schematized in the left panel of Figure 11. There is also, however, a negative correlation between the intervals in the CC condition (left). Although the linear regression is not as steep for CC as it is for CVC, the negative correlation is not predicted by the gestural coordination pattern schematized in the right panel of Figure 11. This indicates that at least some of the tokens lacking a vowel height target, as determined by classification of tongue dorsum height trajectories, retains some aspect the vowel gesture, at least enough to maintain coordination patterns.

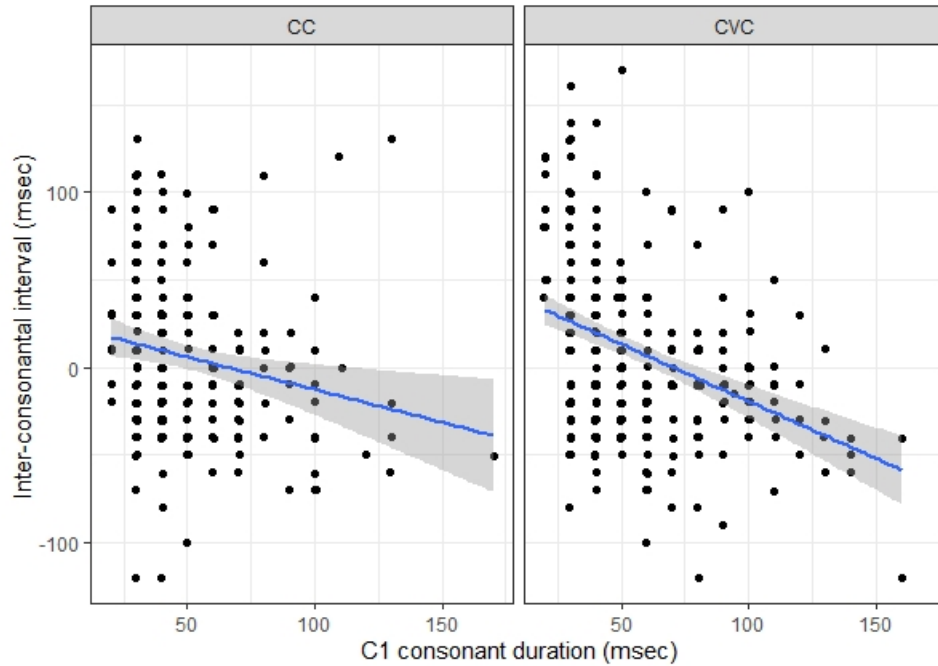


Figure 12: Covariation between ICI and C_1 duration. The left panel=tokens classified as no tongue dorsum raising; the right panel=tokens with a clear tongue dorsum target.

677 To further explore the patterns in Figure 12, we broke down the data by initial conso-
 678 nant and re-plotted as Figure 13. Here, we see that the negative correlation between ICI
 679 and C_1 duration for the CC conditions is only for [ʃ]-initial words. [ɸ]-initial words show
 680 no correlation, the pattern predicted by the coordination topology presented in the right
 681 panel of Figure 11.

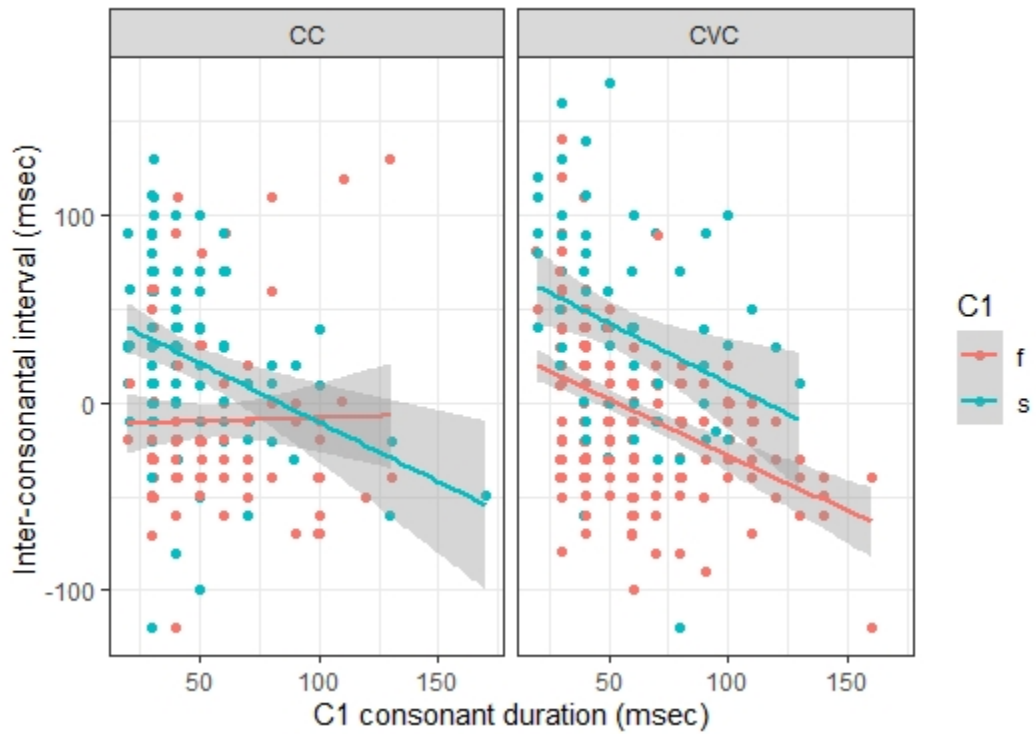


Figure 13: Covariation between ICI and C_1 duration, broken down by initial fricatives. The left panel=tokens classified as no tongue dorsum raising; the right panel=tokens with a clear tongue dorsum target.

682 To assess the statistical significance of the patterns shown in Figure 13, we fit a series
 683 of nested linear mixed effects models to ICI. The models are summarized below.

$$m0 : ICI \sim (1|speaker) + (1|item) \quad (7)$$

$$m1 : ICI \sim C1duration + (1|speaker) + (1|item)$$

$$m2 : ICI \sim C1duration + Cond + (1|speaker) + (1|item)$$

$$m3 : ICI \sim C1duration * Cond + (1|speaker) + (1|item)$$

$$m4 : ICI \sim C1duration * Cond * C1identity + (1|speaker) + (1|item)$$

684 The baseline model included only random effects, a random intercept for speaker and
685 a random intercept for item. To this, we added the following fixed effects: C₁ duration,
686 Cond(ition) (CC vs. CVC), C₁ identity ([ɸ] vs. [j]) and the interactions between these
687 fixed factors. Model comparison is summarized in Table 3. The best model of ICI, as
688 indicated by variance explained and lowest AIC, is the model that contains the three-way
689 interaction between C₁ duration * condition * C₁ identity. This indicates that the effect
690 that C₁ duration has on ICI depends on both condition and C₁ identity, confirming the
691 statistical significance of the differential effect of C₁ duration shown above.

Table 3: Summary of model comparisons.

	Df	AIC	BIC	logLik	deviance	χ^2	χ^2	Df	p
<i>m0</i>	4	5129.9	5146.7	-2560.9	5121.9	—	—	—	—
<i>m1</i>	5	5055.7	5076.6	-2522.8	5045.7	76.2	1.0		< 0.001
<i>m2</i>	6	5046.4	5071.6	-2517.2	5034.4	11.2	1.0		< 0.01
<i>m3</i>	7	5045.0	5074.3	-2515.5	5031.0	3.4	1.0		<i>n.s</i>
<i>m4</i>	11	5027.3	5073.5	-2502.7	5005.3	25.7	4.0		<.001

692 A summary of the best fitting model, *m4*, is provided in Table 4. All three fixed factors
693 were significant, as was the two-way interaction between C₁ duration and condition. The
694 direction of the effects is as expected from the figure above. C₁ duration had a significant
695 negative effect—overall longer C₁ duration predicts shorter ICI ($\beta = -0.39, p < 0.05$).
696 The effect of condition is positive ($\beta = 27.9, p < 0.05$), indicating that tokens in the CVC
697 condition, those classified as having a vowel height target, are substantially longer (ca. 28

ms) than tokens in the CC condition, those classified as lacking a vowel height target. The third fixed factor, C₁ identity, was positive ($\beta = 48.4, p < .001$), indicating that [j]-initial words are longer (ca. 48 ms) than [ϕ]-initial words. Finally, the two-way interaction between C₁ duration and condition was negative ($\beta = -0.36, p < .05$), amplifying the negative effect of C₁ duration in the CVC condition. This pattern of results indicates that the negative effect of C₁ duration on ICI is greatest in the CVC condition.

Table 4: The model summary.

	β	s.e.	df	t	p
(Intercept)	22.3	17.4	10.8	1.3	0.23
C1 duration	-0.4	0.2	474.8	-2.5	0.01
condition_CVC	27.9	11.3	472.7	2.5	0.01
C1_identity_sh	48.4	13.8	131.9	3.5	0.00
C1 duration:Condition_CVC	-0.4	0.2	468.0	-2.0	0.05
C1 duration:C1_identity_sh	-0.3	0.2	477.4	-1.1	0.26
Condition_CVC:C1_identity_sh	-1.4	17.5	467.8	-0.1	0.94
C1 duration:condition_CVC:C1_identity_sh	0.3	0.3	468.0	1.1	0.26

We interpret the above results as indicating that, of the vowels lacking a height target, only those following [ϕ] are completely deleted. It is in this environment that ICI is independent of C₁ duration. Evidence from coordination patterns indicates that the vowels following [j] may retain some target, since variation in [j] duration conditions variation in ICI in a manner similar to CVC sequences.

One possible explanation for this difference between [ϕ] and [j] is related to the predictability of the vowel after these two types of consonants Whang (2018). Due to Japanese phonotactics, only [u] can follow [ϕ], but both [i] and [u] can follow [j]. This may mean that after [j], speakers may need to retain the tongue retraction gesture—or some gesture that would distinguish Japanese [u] from [i]—even when the tongue dorsum raising gesture is absent. This is a post-hoc hypothesis, and needs to be verified in future work.

To summarize, the patterns for [ϕ] replicate what was found in Shaw & Kawahara (2018b), and lends additional credibility to the idea that those vowels that lack a tongue

dorsum height target are likely to be deleted in their entirety, resulting in different gestural coordination patterns between flanking consonants. On the other hand, we have found that vowels following [ʃ] may behave differently: even though the tongue dorsum raising gesture is also deleted in this context, the coordination pattern between flanking consonants is preserved, suggesting that some vestiges of vocalic gestures are retained.

7 Conclusion

Despite extensive past research on high vowel devoicing in Japanese, one issue that has remained open is whether the devoiced vowels are phonologically deleted or not. Following a previous study on this topic (Shaw & Kawahara, 2018b), the current EMA-based experiment explored this question with an extended stimulus set, with a new hypothesis that surrounding consonantal environments may modulate the deletion probability. The current experiment replicated the core findings of Shaw & Kawahara (2018b): in general, devoiced [ɯ̥] either fully retained its articulatory target or entirely lost it, with few intermediate cases, i.e., of only partial reduction. An additional analysis of the covariation between C_1 duration and ICI shows that those tokens that lack tongue dorsum gestures and begin with [ɸ] are likely to be entirely deleted, as they trigger a reorganization of the coordination of flanking consonants. In contrast, tokens that lack tongue dorsum gestures and begin with [ʃ] do not trigger such reorganization, suggesting that they may retain some vestige of the vowel.

Additionally, the current experiment found that different speakers show different sensitivities to the surrounding consonantal environments. These results are of descriptive importance, as we still know very little about the factors that condition variable phonological deletion of devoiced vowels in Japanese or, for that matter, any other languages that exhibit vowel devoicing. The current results highlight the importance of examining such behavior both within- and across- speakers, as sensitivity to phonological factors may also vary within a speech community.

743 **Statement of Ethics**

744 The current experiment was conducted with the approval of Western Sydney University
745 and Keio University (Protocol number: HREC 9482). A consent form was obtained from
746 each participant before the experiment.

747 **Conflict of Interest**

748 The authors declare no conflicts of interest.

749 **Author Contributions Statement**

750 Designing the experiment: JS and SK; data analysis: JS; discussion of the results: JS and
751 SK; writing up the paper: JS and SK.

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